# Serina Chang

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**ACADEMIC POSITIONS**

**University of California, Berkeley**, Berkeley, CA — 2025-present
Assistant Professor, Department of Electrical Engineering and Computer Sciences (50%), UCSF UC Berkeley Joint Program in Computational Precision Health (50%)
Member of the Berkeley AI Research (BAIR) Lab and the Center for Human-Compatible AI (CHAI)

**Microsoft Research**, New York, NY — 2024-2025
Postdoctoral Researcher, Computational Social Science group

**EDUCATION**

**Stanford University**, Stanford, CA — 2019-2024
Ph.D., Computer Science
*Advisors*: Jure Leskovec, Johan Ugander; *Thesis committee*: Dan Jurafsky, Eric Horvitz
*Dissertation*: "Computational Methods for Human Networks and High-Stakes Decisions"

**Columbia University**, New York, NY — 2015-2019
B.A., Computer Science (major), Sociology (concentration), *magna cum laude*
*Advisor*: Kathleen McKeown

**HONORS**

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|---|---|
| Forbes 30 under 30 (United States, Healthcare) | Jan 2026 |
| ACM SIGKDD Dissertation Award | Aug 2025 |
| Google Research Scholar Award | May 2025 |
| Rising Stars in Data Science, University of Chicago and University of California San Diego | Nov 2023 |
| EECS Rising Stars, Georgia Institute of Technology | Nov 2023 |
| Future Faculty Symposium Scholar, Cornell University | Sep 2023 |
| Meta PhD Fellowship, Computational Social Science | Feb 2022 |
| ACM SIGKDD 2021 Best Paper Award, Applied Data Science Track | Aug 2021 |
| Graduate Research Fellowship, National Science Foundation | Sep 2019 |
| The Finch Family Fellowship, Stanford University, School of Engineering | Sep 2019 |
| Phi Beta Kappa, Columbia University | May 2019 |
| Outstanding Undergraduate Researcher Award, Computing Research Association | Jan 2019 |
| Dean's List (all semesters), Columbia University | 2015-2019 |

**RESEARCH INTERESTS**

My research lies at the intersection of **AI and human behavior**, including developing human-centered AI models, evaluating human-AI interactions, simulating and inferring human behaviors with AI, and modeling population behaviors to guide high-stakes public health and policy decisions.

**PAPERS**

*indicates co-first authorship.

**Under Review / Preprint**

[31] Quantifying the Utility of User Simulators for Building Collaborative LLM Assistants
Joseph Suh, Ayush Raj, Minwoo Kang, and **Serina Chang**

[30] Learning Demographic-Conditioned Mobility Trajectories with Aggregate Supervision
Jessie Li, Zhiqing Hong, Toru Shirakawa, and **Serina Chang**

[29] Political Neutrality as Balanced Approval: A Large-Scale Human Evaluation of AI Responses
Jonathan Stray*, David Zhai Yang*, Steven Luo*, Miu Nicole Takagi, and **Serina Chang**

[28] Whose Story Does AI Tell? Investigating LLMs as News Agents
Samia Menon*, Kanav Mittal*, Abigail O'Neill, and **Serina Chang**

[27] Orthogonalized Activation Steering
David Arbour, Erfan Jahanparast, **Serina Chang**, and Avi Feller

[26] Evaluating Large Language Models for Guiding Policy-Level, Multi-Objective Decisions
Ritviksiddha Penchala, **Serina Chang**, and Reza Yaesoubi

[25] ORQA: Occupation Realistic Q&A Benchmark for LLM Professional Assistance
Shreyas Krishnan, **Serina Chang**, and Abhishek Nagaraj

[24] Behavioral AI: Building Algorithms That Understand Us
Keyon Vafa, Ilia Sucholutsky, Katherine M. Collins, Michael S. Bernstein, Andreea Bobu, **Serina Chang**, Diag Davenport, Kawin Ethayarajh, Katy Ilonka Gero, Thomas L. Griffiths, Hoda Heidari, Hannah Rose Kirk, Jon Kleinberg, and Sendhil Mullainathan

[23] Modelling the Public-Health Impact of Indoor Air Quality Interventions on Respiratory Virus Transmission
Adam Howes, Geetha Jeyapragasan, Richard Williamson, David Carel, Harrison Koos, Jacob L. Swett, James Montavon, Vivian Belenky, Paul Lietar, Richard Fitzjohn, Giovanni Charles, **Serina Chang**, Thomas Brewer, and Charles Whittaker

**Published / Accepted**

[22] Same Event, Different Truths: How Next-Generation Social Media Platforms Reframe Political Events
Mao Li, Xinyi Chen, and **Serina Chang**
ICWSM 2027

[21] Graph-Based Alternatives to LLMs for Human Simulation
Joseph Suh, Suhong Moon, and **Serina Chang**
ACL 2026 (main, long paper)

[20] Valid Survey Simulations with Limited Human Data: The Roles of Prompting, Fine-Tuning, and Rectification
Stefan Krsteski, Giuseppe Russo, **Serina Chang**, Robert West, and Kristina Gligorić
ACL 2026 (main, long paper)

[19] What Do Large Language Models Know About Opinions?
Erfan Jahanparast, Zhiqing Hong, and **Serina Chang**
ICLR 2026

[18] Global Approaches to Infectious Disease Surveillance and Modelling
Mark Khurana, Joseph Tsui, Bernardo Gutierrez, Ayush Chopra, Neil Scheidwasser, Harrison Zhu, **Serina Chang**, David Duchêne, Cathal Mills, Rhys Inward, ..., Samir Bhatt, Moritz Kraemer
*Nature Medicine*, 2026

[17] Language Model Fine-Tuning on Scaled Survey Data for Predicting Distributions of Public Opinions
Joseph Suh*, Erfan Jahanparast*, Suhong Moon*, Minwoo Kang*, and **Serina Chang**
ACL 2025 (main, long paper)
Also presented at American Association for Public Opinion Research (AAPOR) Conference 2025 (oral) and the Social Sim and NLPOR Workshops at COLM 2025

[16] ChatBench: From Static Benchmarks to Human-AI Evaluation
**Serina Chang**, Ashton Anderson, and Jake Hofman
ACL 2025 (main, long paper)

[15] LLMs generate structurally realistic social networks but overestimate political homophily
**Serina Chang***, Alicja Chaszczewicz*, Emma Wang, Maya Josifovska, Emma Pierson, and Jure Leskovec
ICWSM 2025
Also presented at IC$^2$S$^2$ 2024 as **Plenary Talk** (2.8% of submissions)

[14] Learning production functions for supply chains with graph neural networks
**Serina Chang**, Zhiyin Lin, Benjamin Yan, Swapnil Bembde, Qi Xiu, Chi Heem Wong, Yu Qin, Frank Kloster, Xi Luo, Raj Palleti, and Jure Leskovec
AAAI 2025 (oral, 4.9% of submissions)
Also presented at Stanford Graph Learning Workshop 2023 and Stanford Causal Science Conference 2023

[13] Artificial intelligence for modelling infectious disease epidemics
Moritz U. G. Kraemer*, Joseph L.-H. Tsui*, **Serina Chang***, Spyros Lytras, Mark P. Khurana, Samantha Vanderslott, Sumali Bajaj, Neil Scheidwasser, Jacob Liam Curran-Sebastian, ..., and Samir Bhatt
*Nature* 2025

[12] Measuring vaccination coverage and concerns of vaccine holdouts from web search logs
**Serina Chang**, Adam Fourney, and Eric Horvitz
*Nature Communications* 2024
Also presented at KDD 2023 Workshop on Epidemiology Meets Data Mining and Knowledge Discovery (oral); KDD 2023 Workshop on Data Science for Social Good (oral); $IC^2S^2$ 2024 (oral)

[11] Inferring dynamic networks from marginals with iterative proportional fitting
**Serina Chang***, Frederic Koehler*, Zhaonan Qu*, Jure Leskovec, and Johan Ugander
ICML 2024
Also presented at *Learning on Graphs* 2023

[10] Estimating geographic spillover effects of COVID-19 policies from large-scale mobility networks
**Serina Chang**, Damir Vrabac, Jure Leskovec, and Johan Ugander
AAAI 2023
Also presented at KDD 2022 Workshop on Data-driven Humanitarian Mapping and Policymaking (oral); $IC^2S^2$ 2022

[9] Computational analysis of 140 years of US political speeches reveals more positive but increasingly polarized framing of immigration
Dallas Card, **Serina Chang**, Chris Becker, Julia Mendelsohn, Rob Voigt, Leah Boustan, Ran Abramitzky, and Dan Jurafsky
*Proceedings of the National Academy of Sciences (PNAS)* 2022

[8] To recommend or not? A model-based comparison of item-matching processes
**Serina Chang** and Johan Ugander
ICWSM 2022 (oral)
Also presented at $IC^2S^2$ 2021 (oral)

[7] Data-driven real-time strategic placement of mobile vaccine distribution sites
Zakaria Mehrab, Mandy Wilson, **Serina Chang**, Galen Harrison, Bryan L. Lewis, Alex Tellionis, Justin Crowe, Dennis Kim, Scott Spillman, Kate Peters, Jure Leskovec, and Madhav Marathe
IAAI 2022

[6] Supporting COVID-19 policy response with large-scale mobility-based modeling
**Serina Chang**, Mandy Wilson, Bryan Lewis, Zakaria Mehrab, Komal K. Dudakiya, Emma Pierson, Pang Wei Koh, Jaline Gerardin, Beth Redbird, David Grusky, Madhav Marathe, and Jure Leskovec
KDD 2021 (oral)
**Best Paper Award**, Applied Data Science Track (1 out of 705 submissions)

[5] Mobility network models of COVID-19 explain inequities and inform reopening
**Serina Chang***, Emma Pierson*, Pang Wei Koh*, Jaline Gerardin, Beth Redbird, David Grusky, and Jure Leskovec
*Nature* 2021
Also presented at Networks 2021 (oral); NeurIPS 2020 COVID-19 Symposium (invited talk); NeurIPS 2020 Machine Learning for Health Workshop
Coverage in 650+ news outlets, including *The New York Times* and *The Washington Post*

[4] Epidemic dynamics in inhomogeneous populations and the role of superspreaders
Kyle Kawagoe*, Mark Rychnovsky*, **Serina Chang**, Greg Huber, Lucy M. Li, Jonathan Miller, Reuven Pnini, Boris Veytsman, and David Yllanes
*Physical Review Research* 2021

[3] Automatically inferring gender associations from language
**Serina Chang** and Kathleen McKeown
EMNLP 2019 (oral, short paper)

[2] Detecting gang-involved escalation on social media using context
**Serina Chang**, Ruiqi Zhong, Ethan Adams, Fei-Tzin Lee, Siddharth Varia, Desmond Patton, William Frey, Chris Kedzie, and Kathleen McKeown
EMNLP 2018 (oral, long paper)

[1] Crowd-sourced iterative annotation for narrative summarization corpora
Jessica Ouyang, **Serina Chang**, and Kathleen McKeown
EACL 2017 (oral, short paper)

**INVITED TALKS**

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| UCLA, Department of Communication, Colloquium Speaker Series | 2027 (upcoming) |
| Iowa State University, Translational AI Center Seminar Series | 2026 (upcoming) |
| INFORMS, Session on AI, Digital Platforms, and Human Attention | 2026 (upcoming) |
| Stanford University, Summer Institute in Computational Social Science (SICSS) | 2026 (upcoming) |
| UC Berkeley, Cognition Colloquium (PSYCH 229A) | 2026 (upcoming) |
| ACL'26 Workshop on NLP and Computational Social Science, Panel on LLM Social Simulation | 2026 |
| Northwestern University, Validating Generative AI-Based Social Sciences | 2026 |
| University of Michigan, Program in Survey and Data Science, Likert Symposium | 2026 |
| Alameda County Public Health Department, All Epi meeting | 2026 |
| UC Berkeley, Berkeley AI Research (BAIR) Lab Seminar | 2026 |
| UC Berkeley, Demography Brown Bag Series | 2026 |
| UC Berkeley, Institute of Transportation Studies Seminar | 2026 |
| UC Berkeley, Applied Statistical Methods (STAT 215B) | 2026 |
| UCSF, Deep Learning for Biological and Clinical Research (BMI/BioE 212) | 2026 |
| UVA Workshop on Social Contagions, Artificial Intelligence, and Democracy (**Keynote**) | 2025 |
| NeurIPS Workshop on LLM Persona Modeling (**Keynote**) | 2025 |
| Conference on AI and Statistics, ISU and National Institute of Statistical Sciences (**Plenary**) | 2025 |
| WHO Pandemic and Epidemic Intelligence Innovation Forum | 2025 |
| UC Berkeley, EECS Colloquium | 2025 |
| EPINEXT: Next-gen Methods for Data-Rich Epidemic Models (satellite of CCS2025) | 2025 |
| UC Berkeley, Biostatistics Seminar | 2025 |
| Frontiers in Computational Health, Berkeley, CA | 2025 |
| CHI 2025 Panels, "Opportunities and Challenges of Applying LLM-Simulated Data to HCI Studies" | 2025 |
| Princeton, Machine Behavior (COS 598B) | 2025 |
| Stanford, AI Agents and Simulations (CS 222) | 2024 |
| MIT, Initiative on Digital Economy (IDE) Seminar | 2024 |
| Learning on Graphs Conference, NYC Meetup | 2024 |
| UC Berkeley, Computational Precision Health Doctoral Seminar | 2024 |
| CMU, AI and Emerging Economies (Course 94-894) | 2024 |
| Cornell Tech, Urban Data (INFO 5430) | 2024 |
| Public Health Insight Podcast | 2024 |
| Oxford, Networks Seminar | 2024 |
| Stanford, Big Data Methods for Behavioral, Social, and Population Health Research (EPI 270) | 2024 |
| CMU School of Computer Science, Machine Learning Department | 2024 |
| UC Berkeley, EECS Department & Computational Precision Health | 2024 |
| University of Illinois Urbana-Champaign, Department of Computer Science | 2024 |
| AAAI'24 Workshop on Graphs and More Complex Structures for Learning and Reasoning | 2024 |
| University of Washington, Paul G. Allen School of Computer Science & Engineering | 2024 |

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| Microsoft Research NYC, Computational Social Science | 2024 |
| Columbia Business School, Decision, Risk, and Operations Division | 2024 |
| Cornell, Departments of Computer Science & Information Science | 2023 |
| Johns Hopkins, Department of Computer Science | 2023 |
| MIT, Department of Political Science & Schwarzman College of Computing | 2023 |
| NYU Stern School of Business, Department of Technology, Operations, and Statistics | 2023 |
| Learning on Graphs Conference, Stanford Meetup | 2023 |
| Northeastern, Network Science Institute Seminar | 2023 |
| Cornell Tech, Applied Data Science: Decision-Making Beyond Prediction (ORIE 5355) | 2023 |
| Stanford, Graph Learning Workshop | 2023 |
| Columbia, Analysis of Networks and Crowds (COMS 6998) | 2023 |
| Cornell Tech, Data Science for Social Change (CS 6382) | 2023 |
| NSF Predictive Intelligence for Pandemic Prevention (PIPP), PandEval Research Team | 2023 |
| NIH National COVID Cohort Collaborative (N3C), Machine Learning Seminar | 2023 |
| Stanford, Algorithmic Fairness Seminar | 2023 |
| Stanford, Networks (MS&E 135) | 2023 |
| Meta, Computational Social Science Seminar | 2023 |
| Stanford, Introduction to Computational Social Science (MS&E 231) | 2022 |
| Stanford, Fundamental Concepts in MS&E (MS&E 302) | 2022 |
| OECD-ODISSEI Webinar on Open Data Infrastructure | 2021 |
| Data Science Connect Conference | 2021 |
| March for Science Podcast | 2021 |
| PathCheck Global Health Innovators Seminar | 2021 |
| Diaries of Social Data Research Podcast | 2021 |
| NeurIPS, COVID-19 Symposium | 2020 |
| AI Science Spotlight Series | 2020 |
| Global Pervasive Computational Epidemiology Seminar | 2020 |
| The Octavian Report Podcast | 2020 |
| Placekey Community Seminar | 2020 |
| Stanford, Stats ML Retreat | 2020 |

## TEACHING

**Instructor**, UC Berkeley, Machine Learning and Human Behavior (CS294-286) — 2025-present
- This is a new graduate-level CS course that I created with projects, paper readings, and paper presentations
- Course material explores the intersection of ML and human behavior, including modeling human behaviors with ML, integrating human feedback into ML algorithms, and societal impacts of large-scale deployed ML systems (e.g., recommender systems)

**Instructor**, UCSF UC Berkeley, Public Health Practicum (CPH201B) — 2026-present
- This is a graduate-level course for PhD students in the Computational Precision Health program
- Course material introduces students to public health research and practice, including lectures from public health experts and in-depth experiences where students conduct 1-1 interviews with public health professionals

**Head course assistant (CA)**, Stanford University, Machine Learning with Graphs (CS224W) — 2021
- Managed team of 9 CAs and class of around 300 students; oversaw lecture slides, assignments, exams, and final projects
- Course material covers the foundations and state-of-the-art of machine learning with graphs, including representation learning, graph neural networks, reasoning over knowledge graphs, and algorithms for large-scale networks

**Instructor**, Girls Who Code, Summer Immersion Program — 2019
- Served as the primary teacher for a classroom of 20 high school girls
- Taught a 7-week curriculum including Python, HTML, CSS, JavaScript, and Arduino

**Instructional assistant**, Columbia University, Data Structures in Java (COMS 3134) — 2017
- Led discussion sections, held weekly office hours, and graded assignments and exams

**Peer tutor**, Columbia University, Computer Science Theory (COMS 3261) — 2017

## ADVISING

**Advising as Faculty**, UC Berkeley

*Current PhD advisees*:

Jessie (Zixin) Li, UCSF-UC Berkeley Computational Precision Health Program — 2026-present

Joseph Suh, UC Berkeley EECS (co-advised with John Canny) — 2024-present

*Visiting PhD students*:

Enock Niyonkuru, PhD student in UCSF BMI, rotation Dec 2025-Mar 2026
Zhiqing Hong, UC Berkeley visiting student, PhD student at Rutgers University Jul-Dec 2025

*Postdoc*:
Bill (Wang) Zhu, UC Berkeley 2026-present

*Undergraduate*: Murari Ganesan, Erfan Jahanparast, Steven Luo, Jonathan Ngai, Ayush Raj, David Zhai Yang (UC Berkeley)

**PhD Committees**
Tienyu (Aaron) Liu, UCSF-UC Berkeley CPH, Qualifying Exam Committee 2026
Rachel Freedman, UC Berkeley EECS, Qualifying Exam Committee 2026
George Ma, UC Berkeley EECS, Qualifying Exam Committee 2026
Noah Baker, UCSF BMI / UCSF-UC Berkeley CPH DE, Qualifying Exam Committee 2026
Suhong Moon, UC Berkeley EECS, Qualifying Exam Committee 2025
Zakaria Mehrab, University of Virginia CS, Dissertation Committee 2025

**Master's Committees**
Kanav Mittal, UC Berkeley EECS 2026
Yi Ju, UC Berkeley EECS 2026

**Research mentees during PhD**, Stanford University
*PhD*: Alicja Chaszczewicz (Stanford CS), Jordan Troutman (Stanford CS)
*Master's*: Damir Vrabac (Stanford CS)
*Undergraduate*: Zhiyin Lin, Daisuke Masuda, Emma Wang, Benjamin Yan (Stanford); Maya Josifovska (UCLA)

**SERVICE**

**Organization**
Co-Organizer, Dagshtul Seminar on "Generative Agent-Based Simulation" 2027
Co-Founder and Co-Organizer, UC Berkeley Seminar on "Population Biology and Ecology of Infectious Diseases" 2026
Co-Organizer, KDD Workshop on Data-driven Decision Making for Public and Population Health 2026
Steering Committee, Center for Human-Compatible AI Workshop 2026
Co-Organizer, NeurIPS Workshop on Behavioral Machine Learning 2024
Co-Organizer, KDD Workshop on Epidemiology meets Data Mining and Knowledge discovery 2024
Program Chair, Machine Learning for Health (ML4H) 2023
Co-Organizer, KDD Workshop on Data Science for Social Good 2023
Panel Moderator, KDD Equity, Diversity & Inclusion (EDI) Day 2023
Faculty Search Committee, Stanford Data Science and School of Engineering 2023
Organizer, NYC Digital Humanities Week, Using Computation to Analyze Gender in Film 2019

**Journal reviewer**
*Science Advances* 2026
*Proceedings of the National Academy of Sciences* (PNAS) 2026
*Nature Communications* 2026
*Proceedings of the National Academy of Sciences* (PNAS) 2025
*The New England Journal of Medicine AI* (NEJM AI) 2024
*Proceedings of the National Academy of Sciences* (PNAS) 2024
*Science Advances* 2023
*Nature Human Behaviour* 2022
*American Journal of Sociology* 2021
*ACM Transactions on Spatial Algorithms and Systems* 2021
*npj Urban Sustainability* 2021

**Conference reviewer**
AAAI, Special Track on AI for Social Impact, Senior Program Committee 2027
ACL Rolling Review 2026
KDD, Research Track, Area Chair 2026
IJCAI-ECAI, AI for Social Good Track, Senior Program Committee (Silver Tier Award) 2026

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| Machine Learning for Health, Area Chair | 2025 |
| AAAI, Special Track on AI for Social Impact, Senior Program Committee | 2025 |
| ACL Rolling Review | 2025 |
| UIST | 2025 |
| WWW | 2025 |
| AAAI | 2025 |
| Machine Learning for Health | 2024 |
| International Conference on Computational Social Science ($IC^2S^2$) | 2024 |
| KDD, epiDAMIK Workshop | 2024 |
| AAAI | 2024 |
| EMNLP | 2024 |
| Machine Learning for Health | 2023 |
| ACL Rolling Review | 2023 |
| ICWSM | 2023 |
| KDD, epiDAMIK Workshop | 2023 |
| KDD, Data Science for Social Good Workshop | 2023 |
| ACL Rolling Review | 2022 |
| Machine Learning for Health | 2022 |
| ACL, NLP for Positive Impact Workshop | 2022 |
| KDD, epiDAMIK Workshop | 2022 |
| ACL-IJCNLP | 2021 |
| Machine Learning for Health | 2021 |
| ACL, NLP for Positive Impact Workshop | 2021 |
| ICLR, AI for Public Health Workshop | 2021 |

**Mentorship**

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| Machine Learning for Health (ML4H), Career Mentorship Program | 2025 |
| IJCNLP-AACL Student Research Workshop, Mentor | 2025 |
| International Conference for Computational Social Science ($IC^2S^2$), Mentor | 2024 |
| Stanford Computer Science, CURIS (undergraduate summer research), Mentor | 2023 |
| Machine Learning for Health (ML4H), Career Mentorship Program | 2023 |
| Stanford Computer Science, Undergraduate Mentoring Program | 2022 |
| Stanford Engineering, Summer Undergraduate Research Fellowship, Mentor | 2021 |

## WORK EXPERIENCE

**AI Advisor**, United Nations Development Programme (UNDP) — 2025-present

- Serving as AI Advisor on “Project Vision: AI Unveils Tomorrow”, which seeks to use AI to analyze how countries are responding to impending risks (e.g., climate change, pandemics, democratic backsliding) and adapting their development strategies

**Consultant**, Honda Research Institute USA — 2025

- Advised on AI-related research projects

**Research Intern**, Microsoft Research — 2022-2024

- Developed graph ML methods to mine search logs and derive public health insights. See Chang, Fourney, and Horvitz, “Measuring vaccination coverage and concerns of vaccine holdouts from web search logs”, *Nature Communications* (2024).

**Software Engineering Intern**, Google, Geo Assistant — 2018

- Built a new, user-facing feature for Google Search and Assistant; implemented changes from backend to frontend

**Engineering Practicum Intern**, Google, Search Site Reliability Engineering (SRE) — 2017

- Improved internal tools for monitoring and tracking requests to Google Now